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Suitability analysis of near South D'Aguilar Nation Park for sustainable development



Yufei He

Shuai (Walker) Feng

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Author Information

Developer

Yufei He (yufei.he@uqconnect.edu.au)
Student ID: s4835200

Conservation

Shuai (Walker) Feng (Shuai.feng@uqconnect.edu.au)
Student ID: s4773988

Introduction

The project provides recommendations and decisions on land use allocation by analyzing the suitability of the target area under different conditions in order to allocate land according to different environmental conditions and land suitability.

The objective area is about 145 square kilometres west of South D’Aguilar National Park. Most of it is farmland. This area contains Brisbane Valley, Warrego Highway and part of the Brisbane River.

Different criteria were selected and suitability tested for each of the two uses to explore the most appropriate land allocation options.



Figure1 Target Area

Suitability analysis is a methodology to identify the suitability of a system (contributors, 2024); in this project, it is land use. The land use suitability analysis needs to convert qualitative conditions to quantitative criteria scores to identify the best area of each objective (Malczewski, 2004). First of all, land use suitability analysis needs an objective. After settling down objectives by surveying the study area, the objectives should be considered in several terms of site conditions. Then, consider how to measure the different site conditions, which results in quantitative suitability criteria. Finally, using GIS to calculate the suitability scores following these quantitative suitability criteria. The workflow of land-use suitability analysis is demonstrated in Figure 2.

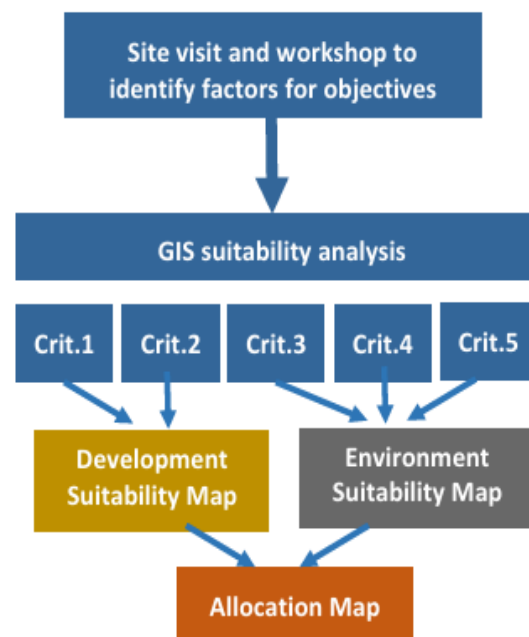


Figure 2: Land-use suitability analysis workflow

Suitability analysis based on GIS has been widely applied in various situations, such as environmental preferences, environmental impact preferences, geographical preferences, and development preferences (Malczewski, 2004). With GIS technology, the land use suitability analysis can have multicriteria, multi-objective and multi-attribute methods.

Methods

The land use suitability analysis includes five procedures: 1) Identify objectives; 2) Detail approaches; 3) Define criteria; 4) Analysis suitability score; 5) Determine land use.

Identifying objectives means defining what will happen in the future and why.

The target area is assumed to be developable and allocated for two uses:

1. Locating 'rural living' area satisfying a country living style and good planning principles
2. Protecting environmental values of the study area and adjoining National Park

Detail approaches mean defining what should be the measurement to determine suitability.

As the developer, to achieve the development objectives, the urban area prefers to be built on flat terrain, closer to the road and river to reduce the cost of transportation and keep away from flooding and potential bushfire impacts.

As the environment protector, we need to identify the species that need to be protected, the extent of protection, and consider areas where fire or human disturbance may occur in order to protect national parks more effectively.

Define criteria means defining criteria for each approach, which uses quantitative value to measure different site conditions. The suitability analysis in this project is based on several standards.

As a developer, the most necessary aspects of development are the geographic conditions, the traffic and the disaster.

As the environment protector, we need to protect organisms and the environment from fire and man-made damage.

Therefore, these criteria define suitability in different ways, which is demonstrated in table 1: Criteria to achieve objective, and the result maps for each criteria are shown in charter Appendix.

Table1 Criteria to achieve objective

Overall Objective	Criteria to achieve objective	How is this assessed spatially	Classify data values to suitability scores to achieve objective					Votes Weighting
			1 Very Low	2 Low	3 Moderate	4 High	5 Very Good	
Rural Living Area	Develop on flat terrain ¹	Slope from DEM	>18°	12°-18°	8°-12°	4°-8°	0°-4°	15
	Lower costs for road access ²	Proximity to roads	>6km	4km-6km	2km-4km	1km-2km	0-1km	45
	River Accessible ³	Lower cost for both water source access and minimise flood impacts	<60m	60m-300m	>10km	3km-10km	300m-3km	25
	Bushfire Risk ⁴	Lower cost for bushfire repairment	Very High Potential Bushfire Intensity	High Potential Bushfire Intensity	Medium Potential Bushfire Intensity	Potential Impact Buffer	None Bushfire Impact	15
Park Protection	Conserve valued vegetation ⁵	Remnant Bio.Status	Non-remnant		No concern		Endangered or of concern	30
	Exclude uses near park ⁶	Buffer park boundary	>500m	300m-500m	200m-300m	100m-200m	0-100m	30
	Probability of bushfire ⁷	Bushfire Hazard Area	Non-impact	Potential impact buffer	Medium potential bushfire intensity	High potential bushfire intensity	Very high potential bushfire intensity	30
	Distance from residential areas ⁸	Buffer residential boundary	0m-2000m	2000m-4000m	4000m-6000m	6000m-8000m	>8000m	10

1. Slope

Development in flat zones is a favourite for residential living, which costs less on infrastructure construction and terrain adaptation (LaGro, 2013). Therefore, the slope scores are divided in Table 1.

2. Transport(road) accessible

Close to transport and other facilities will conform to people's preferences (LaGro, 2013), which makes life more convenient. Therefore, the transport accessible scores are assumed to be those who prefer being close to major roads, as shown in Table 1.

3. River accessible

Close to the river will cost less on water resources. However, being too close to the river will have a high flood impact. According to the government guidance, area within 15 meters of the river is generally forbidden development, area within 60 meters of the river cost heavily on prevent flooding, therefore, area within 60 meters of the river is barely considered as the development area. What's more, the region between the 60 meters to 300 meters is considered as optional area for controlled development; over 50% of the world's population are located at 3km within the river, and only 10% of the population lives more than 10 km away (Kummu et al., 2011). Therefore, the detail of river accessibility scores as shown in Table 1.

4. Bushfire risk

The bushfire only causes damage on both properties and life, therefore, the less potential of bushfire risks the higher suitability of people's preferable. Therefore, the bushfire risk score relates to the bushfire risk extent, which show on Table 1.

5. Remnant Biodiversity Status

The Vegetation Management Act 1999 (Regional Ecosystem Descriptions for Queensland) defines the status of remnant vegetation and its regional representatives within the Biodiversity Framework, and that remnant vegetation is a patch of vegetation consisting of at least 50 per cent undisturbed native species, with a minimum area of 5 hectares.

Biological status was categorised as endangered, of concern and not of concern (Land for Wildlife Queensland, 2011).

To protect remnant vegetation, criteria were developed to assess the biological status of remnant vegetation in the target area based on these three classes, with a biodiversity status of endangered or of concern being assigned five points, organisms of no concern being assigned three points, and non-endangered species being assigned one point.

6. Buffer Park boundary

Developed areas tend to have some impact on neighbouring areas, called edge effects, including weed invasion, physical disturbance of vegetation and soils, incidental litter, and visits from domestic dogs (Smith and Smith, 2010). Direct impacts due to edge effects are within 100m, and buffer distance levels have been established by relevant managers to assess the impact of developed areas on the National Park environment. Therefore, we assigned a score of 5 to the area within the range of 0-100m, followed by 4 and 3 for 100-200m, 200-300m, respectively, 2 for 300-500m, and 1 for the rest of the range.

7. Bushfire Hazard Area

National parks have a lot of natural vegetation that fuels fires, making national parks very vulnerable to bushfires. Effective management of national parks can prevent or mitigate exposure to bushfires by establishing firebreaks, detecting weather conditions or developing strategies to deal with fires, among other things (Gill & Stephens, 2009). Therefore, we can categorise and assign a value to the bushfire risk zones in the target area, where the greater the likelihood of bushfires, the higher the score, and the more important it is to take measures to prevent them.

8. Distance from residential areas

High-intensity human activities in residential areas often have an impact on national parks, such as pollution, habitat destruction and impacts on wildlife, etc., so increasing the distance from residential areas can help to reduce these impacts and contribute to better environmental protection (Hockings et al., 2006). Creating buffer zones around national parks also reduces the environmental impacts of nearby residents, and the further the distance from residential areas, the better it is for protecting the park's natural environment.

Analysis suitability score by following the criteria just defined will generate the suitability score of each area, which quantifies the suitability.

According to the criteria of development, the area development suitability score equation is:

Development suitability

$$= 0.15 * Slop + 0.45 * transport\ accessible + 0.25 * river\ accessible + 0.15 * bushfire\ risk$$

According to the criteria of conservation, the environment protection suitability score equation is:

Conservation suitability

$$= 0.3 * Remnant\ Biodiversity + 0.3 * Park\ Buffer + 0.3 * Bushfire\ Suitability + 0.1 * Residential\ Suitability$$

Figure 3 and Figure 4 demonstrate the results of Development suitability scores and Conservation suitability scores respectively:

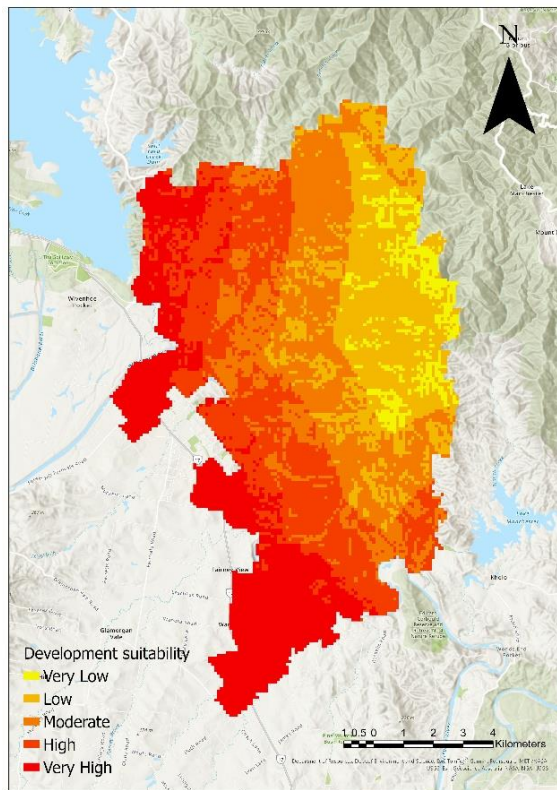


Figure 3 Development suitability Map

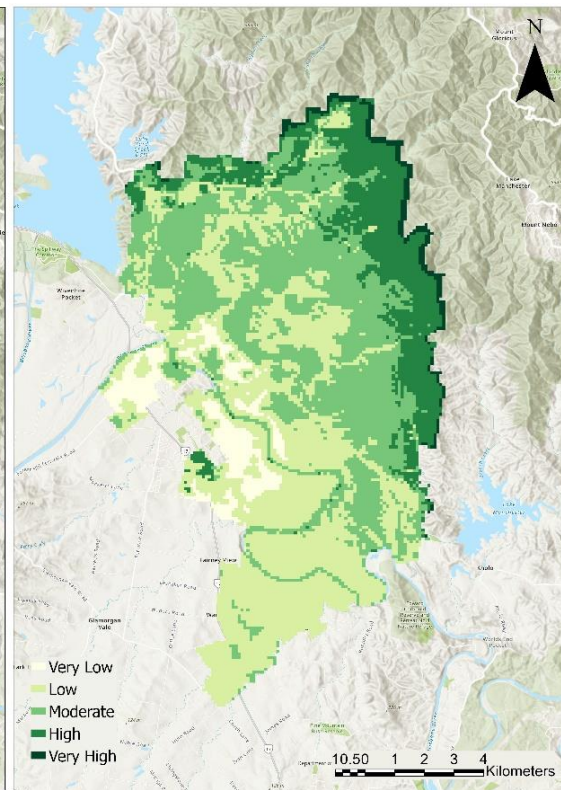


Figure 4 Conservation suitability Map

Determine land use should analyze the suitability criteria results and generate the most perfect area of the objective. This procedure is significant, especially when the suitability criteria are multi-objective.

For areas with high scores on both objectives (both scores of 4 and above), they were allocated to Development Areas due to their small size and spread out away from the National Park and in line with the allocation of surrounding areas.



Figure 5 Areas with high scores on both objectives

Areas with a score of 3 on the park protection side and a low score on the development side are allocated for the protection of national parks due to their proximity to the national parks and their high impact on the park environment, which shown on figure 6 as Area Allocation.



OID	Value	Count	aoi_devsui	int_overal	AreaKm	Allocation
3	4	2094	2	3	20.94	Environment
4	5	2106	1	3	21.06	Environment

Figure 6 Area Allocation



OID	Value	Count	aoi_devsui	int_overal	AreaKm	Allocation
7	8	963	3	3	9.63	Development
10	11	931	3	2	9.31	Development
15	16	1624	3	1	16.24	Development

Figure 7 Area Allocation

For areas with a score of 3 on the development side and a low score on the protection side of the park, they were allocated for development purposes due to their distribution farther away from the national park, their lesser environmental impact on the park and their proximity to residential areas, which shown on figure 7 as Area Allocation.

Results

The figure below shows a plot of the data for the allocation of land to the target area, with the horizontal coordinate being the score for the development area and the vertical coordinate being the score for the protection of the national park, with the orange dots being the area allocated for the development area, the green dots being the area allocated for the protection of the national park, and the grey dots being the unallocated area.

It is worth noting that the residual vegetation criterion used by the park protectors, in which the residual vegetation is mostly distributed along the river rather than around the national park, and the higher scores near the park in the other criteria, resulted in a maximum score of only 4.4 in the resultant image, with no areas close to a score of 5.

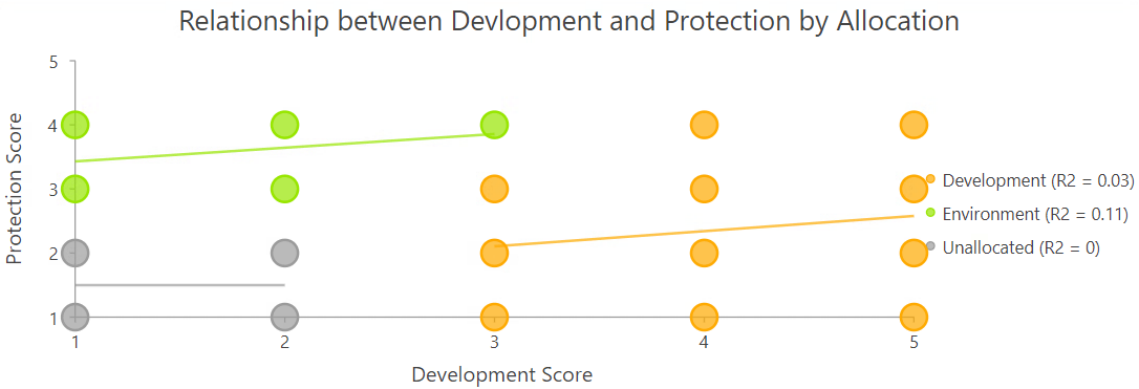


Figure 8 Relationship between Development and Protection by Allocation

Figure 9 demonstrates the final allocation results.

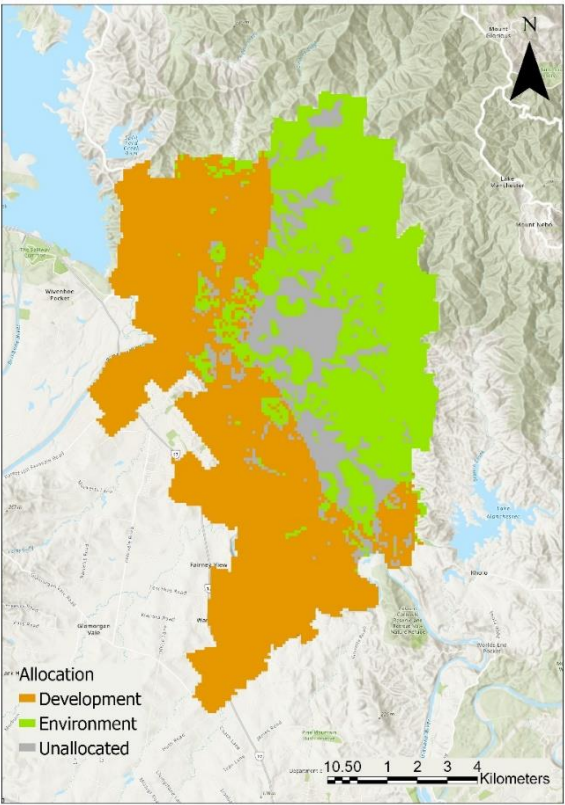


Figure 9 The allocation map

Conclusion

The project establishes different types of criteria to advise on the allocation of land in the target area from the perspectives of both national park protectors and urban developers. For each criterion, a spatial analysis tool is used to statistically calculate the target area and produce the corresponding result maps, which are then superimposed on each other before the land is allocated by manual decision-making methods.

The land on the protection side of the national park will be used for establishing environmental protection facilities or directly for maintaining the ecological environment, while the land on the development side will be used for building and developing rural living areas.

Appendixes

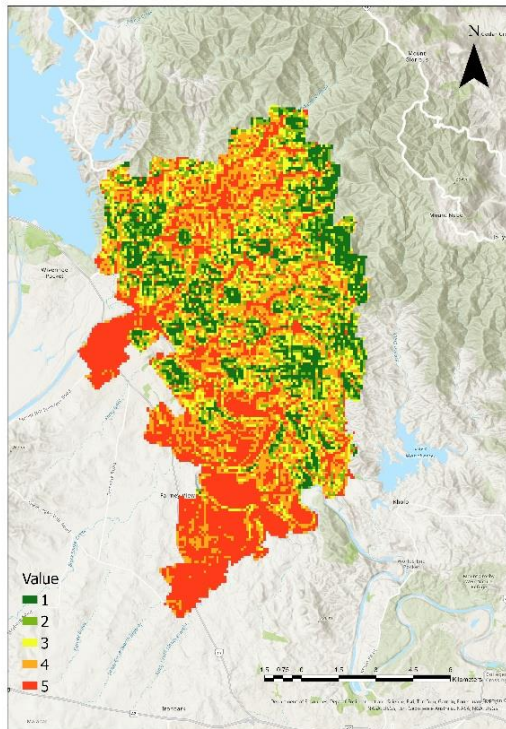


Figure 10 Slope Suitability Map

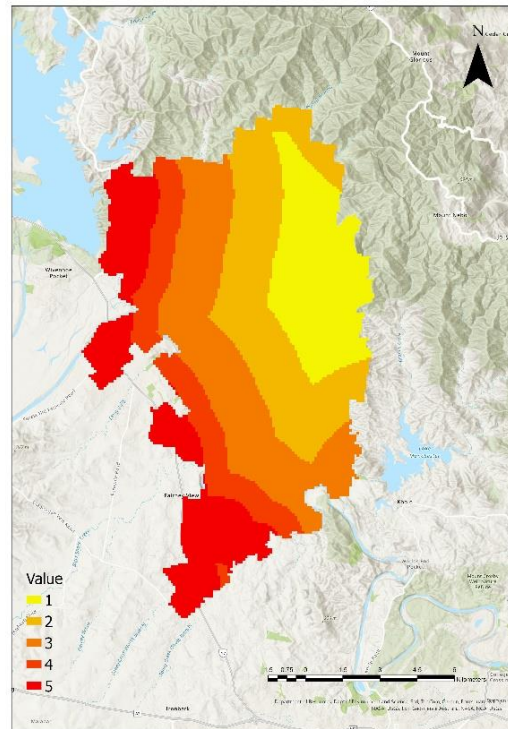


Figure 11 Transport Suitability Map

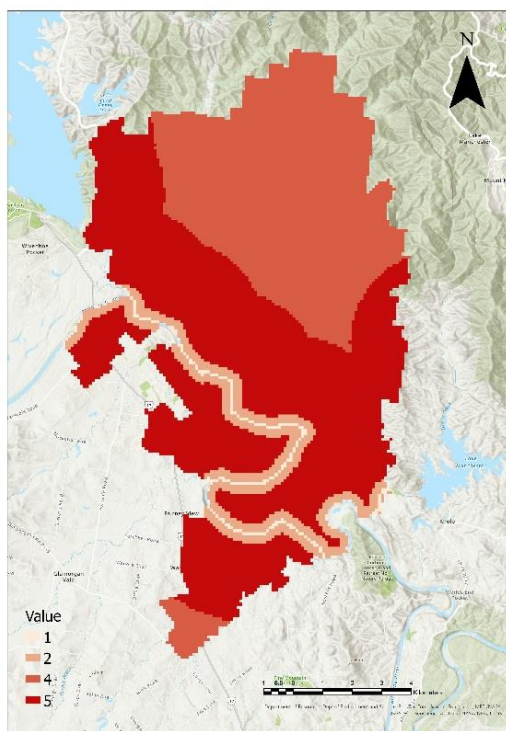


Figure 12 River Suitability Map

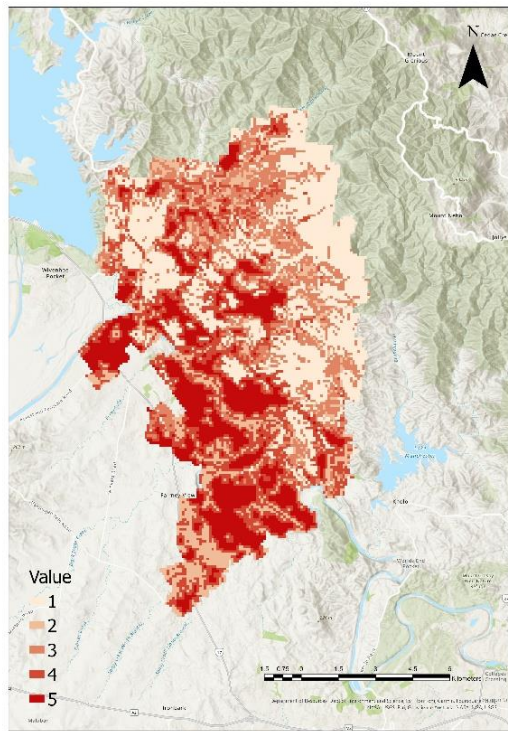


Figure 13 Bushfire Suitability Map

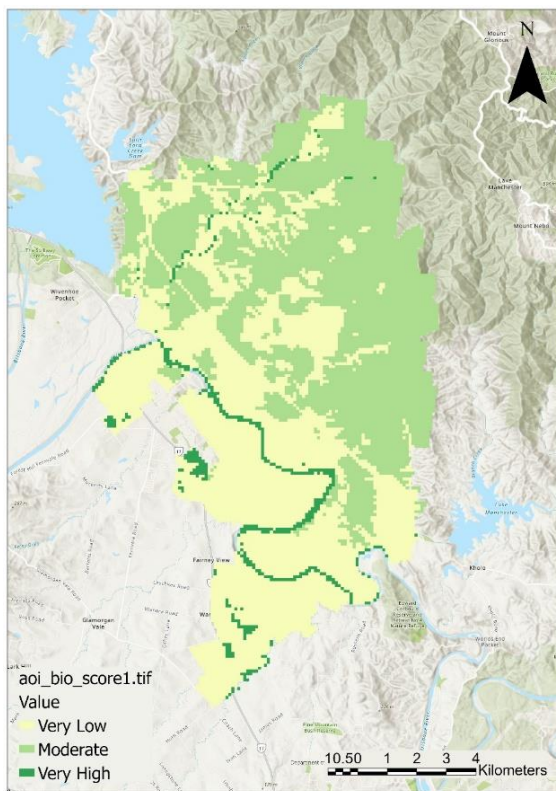


Figure 14 Remnant Biodiversity Suitability Map

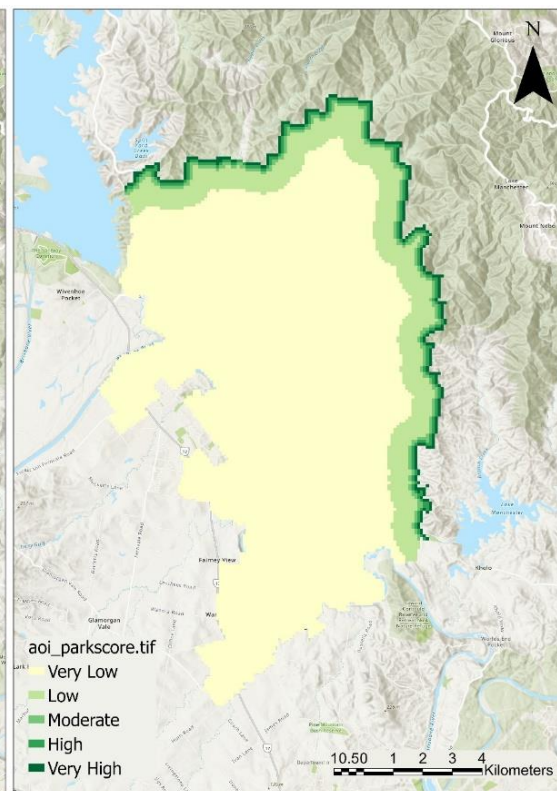


Figure 15 Park Suitability Map

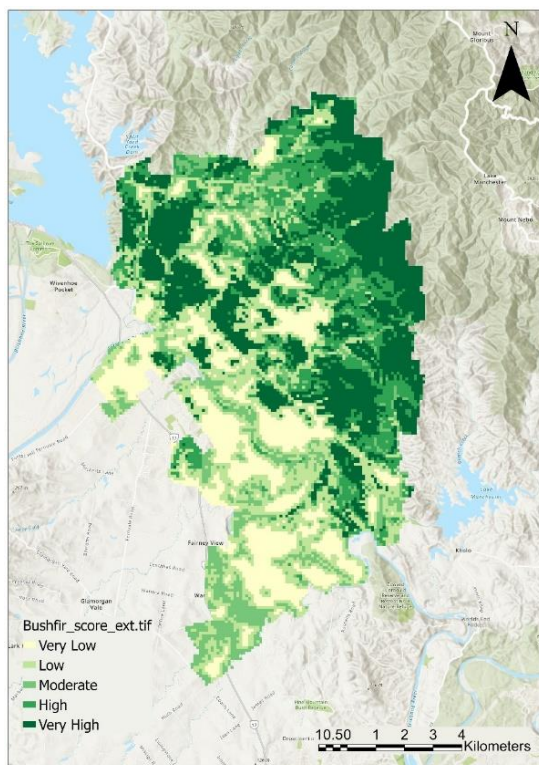


Figure 16 Bushfire Suitability Map

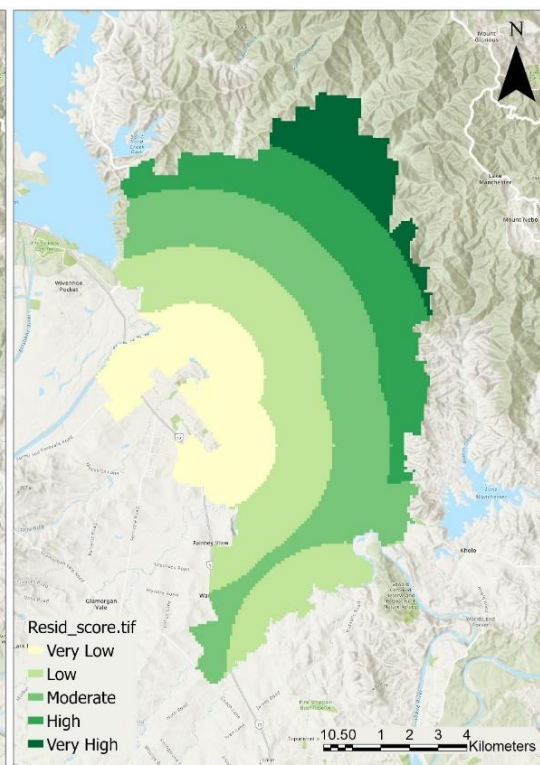


Figure 17 Residential Suitability Map

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